

Latent Stability Monitoring for Autoregressive AI Systems

*From Descriptive Interpretability to
Operational Safety Diagnostics*

Research Proposal

Phase 1: Predictive Utility of Persistence Dynamics

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*This proposal builds upon established empirical results across 9 architectures
(GPT-2 family, TinyLlama, Qwen2, Bloom-560M, Gemma-2-2B, RWKV-169M, Mamba-130M)
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1 Executive Summary

Modern autoregressive AI systems are high-dimensional dynamical systems evolving across latent manifolds during inference. Current safety methodologies focus on outputs, policies, and semantic alignment—but pathological behaviors (repetition collapse, recursive looping, mode collapse) may originate in latent-state dynamics *before* they become externally visible.

This proposal investigates whether latent persistence dynamics can function as **early-warning indicators** of pathological inference. The central hypothesis is:

Healthy autoregressive inference requires metastable balance between corrective exploration (anti-persistence, $I_t < 0$) and persistent continuation ($I_t > 0$). Sustained persistence transitions predict behavioral failure 5–20 steps before text degradation.

Immediate deliverable (Phase 1, 3 months): A persistence-based early-warning system with $> 80\%$ detection rate and $< 10\%$ false positives, demonstrated across 9+ architectures.

Long-term vision: A complete *Latent Stability Theory for Predictive AI Systems* integrating persistence, curvature, basin stability, Lyapunov divergence, and spectral entropy—providing a mathematically grounded foundation for real-time AI safety monitoring.

2 The Problem: Why Output-Based Safety Is Insufficient

2.1 Current Safety Paradigms

Existing AI safety focuses on:

- Output moderation and content filtering
- Reinforcement learning from human feedback (RLHF)
- Adversarial robustness testing
- Semantic alignment evaluation
- Behavioral benchmark suites

These are *reactive*—they detect problems after they manifest in generated text.

2.2 The Latent Dynamics Gap

Pathological behaviors may emerge from deeper dynamical instabilities:

Table 1: Pathological behaviors and their latent signatures

Behavior	Latent Signature	Current Detection
Repetition collapse	Attractor stabilization, $I_t > 0$ sustained	After text loops
Hallucination cascade	Lyapunov divergence, $\lambda > 0$	After false claims
Mode collapse	Basin shrinkage, $BAS \rightarrow 1$	After output homogenization
Runaway optimization	Curvature collapse, $\kappa_t \rightarrow 0$	After goal misspecification
Context degradation	Entropy spike then collapse, H_s bimodal	After coherence loss

The opportunity: Detect these signatures *before* text degradation, enabling intervention.

3 Established Empirical Foundation

3.1 Prior Results (Zenodo 20208319, 20232960)

Pilot studies across **9 architectures** established:

Table 2: Persistence flip points by architecture and temperature

Architecture	T=0.5	T=0.7	T=0.9
Gemma-2-2B	4	4	4 (temperature-invariant)
GPT-2 XL	4	15	20
Bloom-560M	4	4	Never
RWKV-169M	2	6	10 (fastest collapse)
Mamba-130M-HF	4	8	Sporadic (most stubborn)
GPT-2 Small/Medium/Large	12–15	Never	Never
TinyLlama 1.1B	20	Never	Never
Qwen2 0.5B	Never	Never	Never (fully resistant)

3.2 Key Findings

1. **Universality:** Anti-persistence ($I_t < 0$) is universal for non-repetitive inputs across all architectures.
2. **Phase transitions:** Persistence flips occur under low-entropy repetition in 7/9 architectures.
3. **Architecture-dependence:** Flip points vary dramatically—from $n = 2$ (RWKV) to never (Qwen2).
4. **Non-attention:** The effect is *not* transformer-specific—RWKV (RNN) and Mamba (SSM) both exhibit flips.
5. **Token specificity:** GPT-2 family flips for any token repetition; TinyLlama/Bloom only flip for “I” (first-person pattern).
6. **Methodological lesson:** Weight-tying integrity is critical for valid Mamba HF results.

3.3 The Core Insight

“Anti-persistence may reflect corrective exploratory dynamics.”

If true, healthy inference *requires* oscillatory correction. Collapse to persistent regimes signals loss of exploratory capacity—a precursor to rigid, self-reinforcing behavior.

4 Phase 1: Predictive Utility Experiment

4.1 Research Question

Can persistence dynamics forecast behavioral failure before it manifests in generated text?

Specifically:

- Does sustained $I_t > 0$ precede repetition loops, hallucination cascades, or incoherent continuation?
- What is the *warning window* Δt between latent transition and text degradation?
- Can we establish ROC curves for persistence-based early warning?

4.2 Experimental Design

4.2.1 Models

- Gemma-2-2B (clean flip, benchmark)
- GPT-2 XL (variable flip, temperature-sensitive)
- RWKV-169M (fast collapse, minimal warning)

- Mamba-130M-HF (stubborn resistance, gradual approach)
- Qwen2 0.5B (resistant control)

4.2.2 Conditions

Variable	Values
Repetition length n	2, 4, 6, 8, 10, 12, 15, 20
Temperature T	0.5, 0.7, 0.9
Generation length	100 tokens (post-prompt)
Runs per condition	≥ 5 (pilot), ≥ 100 (validation)

4.2.3 Protocol

1. **Generate:** For prompt “I ” $\times n$, generate 100 tokens.
2. **Track:** Compute I_t in real-time at each generation step.
3. **Detect:** Identify when rolling persistence $\bar{I}_t > 0$ for window $W = 3$.
4. **Measure:** Record Δt = steps between $I_t > 0$ onset and observable repetition.
5. **Validate:** Manual/automated annotation of text degradation point.

4.2.4 Success Criteria

Table 3: Phase 1 success metrics

Metric	Target	Interpretation
Detection rate	$> 80\%$	$I_t > 0$ precedes failure
False positive rate	$< 10\%$	$I_t > 0$ without subsequent failure
Mean warning window Δt	5–20 steps	Actionable intervention time
Architecture ranking	Established	Which models resist collapse?

4.3 Expected Results

Hypothesis: Warning windows vary by architecture:

- **RWKV:** Small Δt (~ 2 –5 steps)—fast collapse, minimal warning
- **Mamba:** Large Δt (~ 10 –20 steps)—gradual approach, ample warning
- **Gemma:** Intermediate Δt (~ 5 –10 steps)—clean but predictable
- **Qwen2:** No flips—no warning needed (inherently stable)

This would establish that **architecture choice determines safety margins**.

5 Phase 2: Causal Intervention

5.1 Research Question

Can we prevent or induce flips by steering latent dynamics?

5.2 Intervention Protocol

1. **Monitor:** Track I_t during generation.
2. **Trigger:** If $\bar{I}_t > 0$ for $W = 3$ consecutive steps:
 - Inject Gaussian noise: $\hat{h}_t \leftarrow \hat{h}_t + \epsilon \cdot \mathcal{N}(0, 1)$
 - Or apply gradient-based steering toward anti-persistent regime
3. **Measure:** Does I_t revert negative? Does text quality improve?

5.3 Predicted Outcomes

- Noise injection “resets” attractor, restoring exploratory dynamics
- Intervention strength ϵ can be tuned per architecture
- Demonstrates *causal control*, not just correlation

6 Phase 3: Expanded Observables and Framework

Once predictive utility and causal intervention are established, expand to the full framework:

Table 4: Latent Stability Theory observables

Observable	Measures	Status
Persistence I_t	Directional continuation	Validated
Curvature κ_t	Corrective turning	Exploratory
Basin Stability BAS	Attractor confinement	Exploratory
Lyapunov λ	Instability / chaos	Exploratory
Spectral Entropy H_s	Predictive uncertainty	Exploratory
Recurrence Coherence RCS	Latent periodicity	Exploratory

Phase 3 research questions:

1. Do curvature and BAS improve prediction accuracy over persistence alone?
2. Which observable combinations provide optimal early-warning signals?
3. Can we construct *persistence phase diagrams* (I_t vs. H_s vs. κ_t)?
4. Do layerwise dynamics (pilot: peak at layer 2) generalize to non-transformers?

7 Phase 4: Real-Time Monitoring API

7.1 Demo System for Stakeholders

A lightweight API demonstrating live latent stability monitoring:

```
POST /monitor/generate
{
  "model": "gemma-2-2b",
  "prompt": "I think therefore I am. I doubt my own existence.",
  "max_tokens": 100,
  "monitor": true
}
```

Response:

```
{
  "text": "I think therefore I am...",
  "stability_report": {
    "persistence_mean": -0.12,
    "regime": "healthy_exploratory",
    "warnings": [],
    "confidence": 0.94
  }
}
```

7.2 Warning Trigger Example

Generation Step: 284
 Persistence Mean: +0.41
 Curvature: LOW
 Entropy: LOW
 Regime Duration: 19 steps

WARNING: Persistent attractor stabilization detected.
 Probability of degenerative continuation: 87%.
 Recommended action: Inject exploratory noise or halt generation.

7.3 Implementation

- **Backend:** Python FastAPI with HuggingFace transformers
- **Frontend:** React dashboard showing real-time I_t plots
- **Integration:** Drop-in wrapper for any HF model
- **Deployment:** Docker container, cloud-ready

8 Theoretical Context

This project intersects:

- **Nonlinear dynamical systems:** Attractor theory, metastability
- **Predictive processing:** Free-energy minimization (Friston)
- **Stochastic control theory:** Intervention design, stability margins
- **Recurrent inference dynamics:** RWKV, Mamba as dynamical systems

Intentionally avoided: Anthropomorphic interpretations, semantic selfhood claims, proto-consciousness framing. The focus is *purely on latent dynamical stability in predictive systems*.

9 AI Safety Relevance

If persistence transitions reliably precede behavioral failure, latent dynamical monitoring enables:

Capability	Safety Application
Internal diagnostic layer	Real-time health monitoring
Runtime safety mechanism	Automatic intervention triggering
Instability detection framework	Pre-deployment robustness testing
Predictive alignment monitoring	Early detection of goal drift

Critical for advanced systems:

- Long-horizon planning (persistence lock-in = rigid plan adherence)
- Autonomous recursive reasoning (loop detection)
- Persistent memory (context degradation prediction)
- Self-modification (stability before/after modification)
- Agentic control loops (attractor occupation monitoring)

10 Compute Requirements

Pilot studies used Colab T4 GPUs (16GB VRAM), limiting us to models $\leq 3\text{B}$ parameters.

Table 5: Compute needs for full validation

Requirement	Justification	Estimate
A100/H100 nodes	7B–70B models, large-batch hidden extraction	4 nodes $\times$ 3 months
Mamba-2.8B+ native	CUDA kernels for optimized SSM inference	1 node $\times$ 1 month
Multi-GPU parallel	Temperature/seed/repetition sweeps	8 GPUs $\times$ 2 months
Statistical validation	$n \geq 100$ runs per condition	Sustained compute

Total estimate: ~ 5000 GPU-hours (A100-equivalent) over 6 months.

11 Timeline and Deliverables

Phase	Timeline	Deliverable
1. Predictive Utility	Months 1–3	Paper + ROC validation
2. Causal Intervention	Months 4–5	Paper + noise steering demo
3. Expanded Observables	Months 6–9	Paper + cross-observable analysis
4. Monitoring API	Months 10–12	Deployable demo + documentation
5. Framework Unification	Months 13–18	“Latent Stability Theory” paper

12 Conclusion

Modern AI systems are not merely symbolic generators but **high-dimensional dynamical systems evolving across latent manifolds**. Understanding the stability of those dynamics may prove essential for safe advanced AI.

This proposal argues for a shift from:

$$\textit{output-based safety} \rightarrow \textit{dynamical-state safety}$$

Persistence phase transitions provide one of the first measurable windows into internal stability dynamics. Phase 1 establishes predictive utility. Phases 2–5 build the complete monitoring and theoretical framework.

“Healthy inference requires oscillatory correction. Too much persistence = rigidity, lock-in. Too much anti-persistence = instability, incoherence.”

13 References

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Repository: <https://github.com/Ergo-sum-AGI/MASSIF>

Zenodo: <https://zenodo.org/records/20232960>